

Adroit Technologies

Protocol Driver

Name	Spectrum Tele-RANGER
DLL	SPECTR



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Contents

1. Overview	4
2. Wiring	5
3. Device Configuration	7
3.1. Device Configuration	7
3.1.1. Port Settings (Device Specific)	7
3.1.2. Central Station Details (Global affects all devices)	8
3.1.3. Outstation Logging "GL" Functions (Device Specific)	8
3.1.4. Outstation Modem Identification Strings (Device Specific)	8
3.1.5. Outstation Details (Device Specific)	9
3.1.6. Outstation Checksum Generation (Device Specific)	9
3.2. Modem Dialog	10
3.3. Master/Standby	11
4. Supported Addresses	12
4.1. Modem Configuration Strings (x2 command set)	12
4.1.1. Example Connection String:	12
4.2. Basic Structure of Driver	13
4.3. Log Retrieval	13
4.4. Addressing Notation	14
4.5. TeleRanger Module Types and Commands	16
5. Protocol Definition and Considerations	17
5.1. Basic Telegram Format	17
5.2.	17
5.3. Fields	17
5.3.1. Command Type Breakdown	17
5.3.2. Command Words	18
5.4. Datascope Display	19
5.5. Advanced Registry Features: (rev 41 and above)	19
6. General Notes and Troubleshooting	20
6.1. Contact Details	20
6.2. Adroit Technologies Knowledge Base	20
7. Appendix A – How to Troubleshoot and Setup Modems	21
7.1. 1 st Steps	21
7.2. For Remote Modem	21
7.3. For Master Modem	21
8. Revisions	22
8.1. Document Revision	22
8.2. DLL Revision	22

1. Overview

Adroit supports communications with the Spectrum TeleRanger Radio Telemetry System, using the Protocol Driver SPECTR.DLL. Related documents that may be read in conjunction with this document are:

- Engineering Note EN-5 SCADA Protocol Serial Interface for the tele-RANGER Radio Telemetry System

2. Wiring

PLC 25 pin		PC 25 pin
1	— Shield	1
2	————	3
3	————	2
4	□ □	4
5		5
6	□ □	6
8		8
20		20
7	————	7



PLEASE NOTE

Connect shield to PLC side only

Master side: DTE->DCE "Modem cable"

Use a standard modem cable 1 to 1, 2 to 2 ... 9 to 9. Usually supplied with your modem.

Remote side: TeleFlex/Micro DTE->DCE "Modem cable"

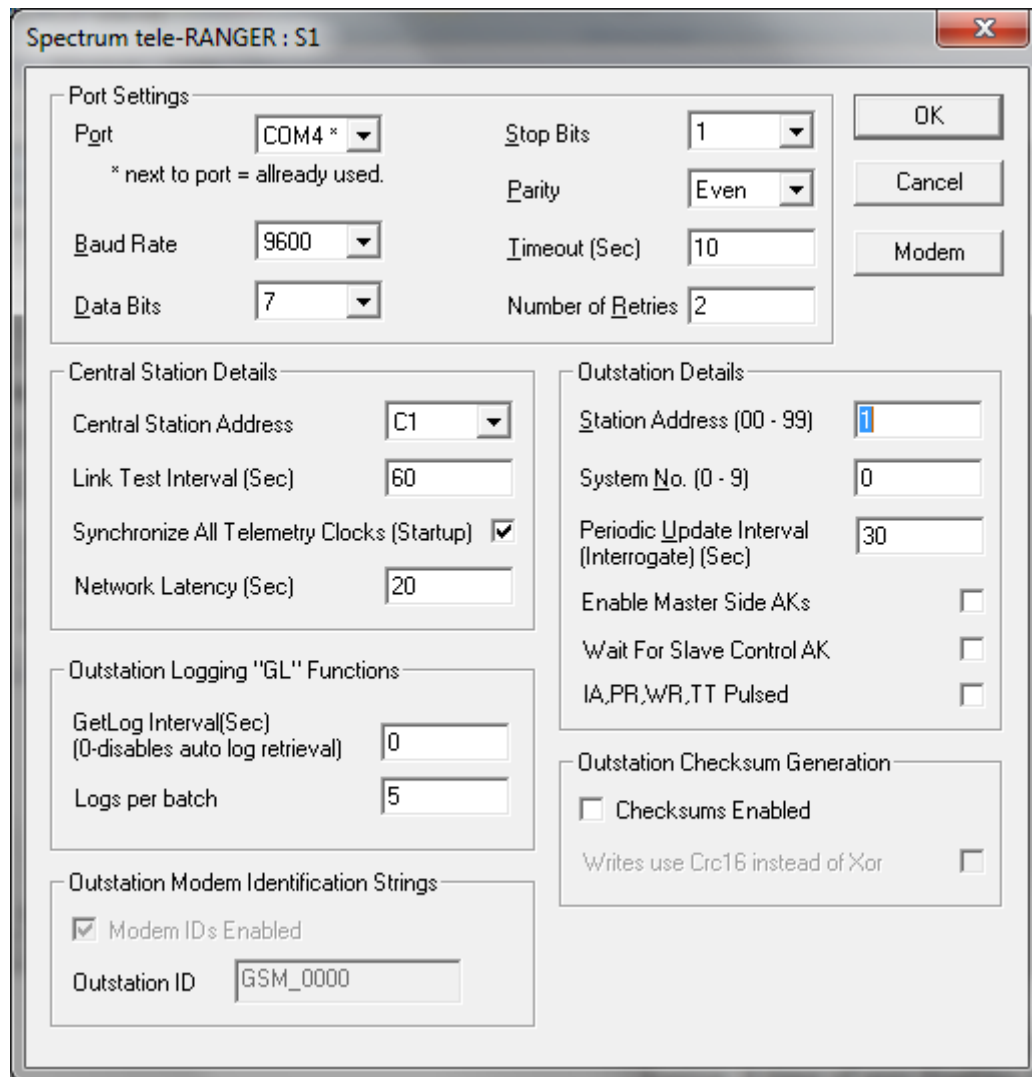
PLC 9 pin		PC 9 pin
1		1
2	————	2
3	————	3
4		6
5	————	5
6		4
7	———— □	7
8		8
9		9

TeleFlex/Micro DTE->DTE "Programming /Comms cable"

PLC 9 pin		PC 9 pin
1		1
2	————	3
3	————	2
4		4
5	————	5
6		6
7	————	8
8	————	7
9		9

3. Device Configuration

3.1. Device Configuration



3.1.1. Port Settings (Device Specific)

Port configuration for the master.

Port - This field defines the communications port used to connect *Adroit* and the teleRanger RS-232 module - COM1, COM2, COM3, etc.

Baud Rate - The baud rate of the RS-232 serial interface, one of: 300, 600, 1200, 2400, 4800, 9600, 19200

Data Bits - The data bits required by the comms port. (7/*8)

Stop Bits - The Stop bits required by the comms port. (*1/2)

Parity - ????

Timeout (Sec) - The time (in seconds) that a response will be waited for, before retrying or timing out

Number of Retries - Number of times a link test (or other command) will be sent without receiving the corresponding response, before a communications failure is flagged. Not inclusive

3.1.2. Central Station Details (Global affects all devices)

Items configured in this group are common to all outstations using the same serial port. What this means is if you want to change a parameter such as, say, Link Test Interval, you will need to change it for all other devices on that particular communications resource (com port), if not then a new port should be used, which implies a different teleRanger RS-232 module, etc.

Please Note non device specific messages like SM and LT etc will not have a checksum at all, the spectrum device will not expect one either.

Central Station Address - The teleRanger central station address, C1 -> C2

Link Test Interval (Sec) - The time (in seconds) between the sending of successive link test ("LT") commands from Adroit to the teleRanger central station. A time of zero disables link testing

Synchronize All Telemetry Clocks (Startup) - When active, this check box will cause all outstations to be time synchronized with the SCADA system clock, This will happen once when the driver loads. Manual syncs via OSCTL.5 will need to be done when in modem mode ... OSHDR.16 can be driven via an Expression Agent.

Time between "IR" messages The number of seconds to leave between any two stations performing an "IR" or interrogate command.

Latency (Sec) - The minimum time delay (in seconds) between the sending of ANY (IR or GL) request commands from Adroit to the outstation. Any one will reset the timer to force the other to delay. This is NOT global it is only a delay between two IR/GL polls to the SAME station. Manual commands (IR and GL) are unaffected and will go out as normal.

3.1.3. Outstation Logging "GL" Functions (Device Specific)

GetLog Interval (Sec) - Default is 0 (disabled). This is a timer to generate an automatic GL request to a device every x Seconds. If set to 0 the timer is disabled and GL's will have to be done from within adroit either via osctl.7 These GL messages obey the *OQD*.

Logs per batch - Default is 5. This is the number of logs to request either in autolog mode or via OSCTL.7 bit.

3.1.4. Outstation Modem Identification Strings (Device Specific)

Modem ID's Enabled - Default : Checked, Used for backward compatibility to older tele-ranger equipment where no ID's are available and where unsolicited mode is unavailable too.
(ONLY Used In Modem MODE!)

Outstation ID - This is the handshake string required to complete a modem connection ... It just adds an extra layer of identification between master and slaves.
Just type in the string specified in the RTU csv file eg.GSM_7778
(ONLY Used In Modem MODE!)

3.1.5. Outstation Details (Device Specific)

In contrast with the Global Options section, each teleRanger outstation may have a different set of outstation details. Generally, the System No. will be common to all outstations, but each outstation will have a unique Station Address.

Station Address - The teleRanger outstation number ("01 to "99")

System No. - The teleRanger system number ("0" to "9") that this outstation belongs to

Periodic Update (sec) (Interrogate) Interval - You will be forced to select 0 this when in modem mode, manual polling will be required in modem mode. The time (in seconds) between the sending of successive update requests ("IR") commands from Adroit to this outstation. This is NOT global it only a delay between two IR polls to the SAME station.

Enable Master Side AKs - Default unchecked. When checked, upon reception of any messages of type Change AC/DC etc, Adroit will transmit an AK Acknowledgement back to the originating outstation.

Wait For Slave Control AK - Default is unchecked. When checked All controls are checked for acknowledgement, i.e. , no new controls will be sent until the pending/current control is completed via reception of the ack. This is per device, therefore if an ack for and IA is "pending", no other control will be sent until the IA-Ack is received or the IA timed-out. This will prevent multiple controls being sent, but the multiple controls will however be queued for transmission later. Controls will also be retried via the timeout mechanism.

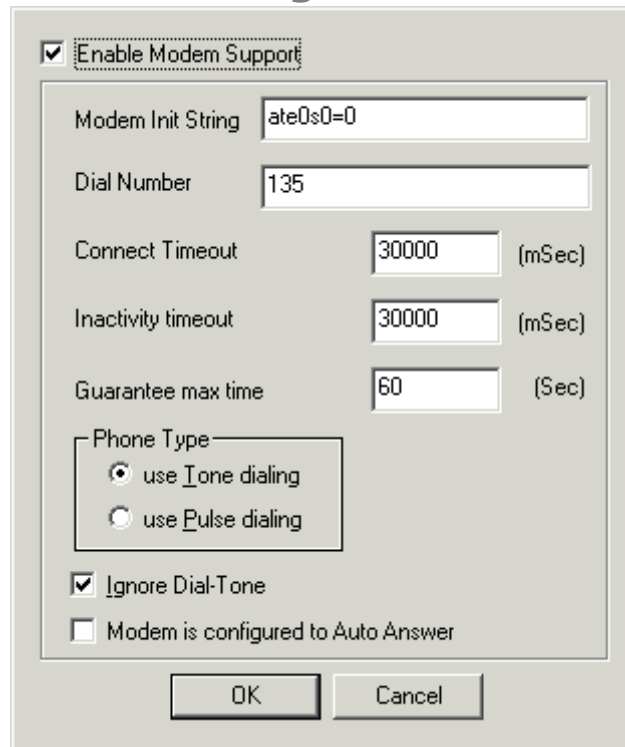
IA/IR/WR/TT Pulsed - Default is unchecked. When checked OSHDR.11/12/13 are pulsed on then off immediately again (useful for triggers).

3.1.6. Outstation Checksum Generation (Device Specific)

Checksums Enabled - Default is unchecked. When checked Checksums will be used by Adroit either CRC or XOR depending on selection below. These are the outgoing master messages; the driver will automatically accept any type of check summing coming from the outstation regardless of these settings.

OutGoing messages use CRC16 instead of XOR Default is unchecked. When checked all outgoing messages will use CRC16 check summing rather than XOR when unchecked.

3.2. Modem Dialog



The image shows a 'Modem Dialog' window. At the top, there is a checkbox labeled 'Enable Modem Support' which is checked. Below this, there are several input fields: 'Modem Init String' with the value 'ate0s0=0', 'Dial Number' with the value '135', 'Connect Timeout' with the value '30000' (mSec), 'Inactivity timeout' with the value '30000' (mSec), and 'Guarantee max time' with the value '60' (Sec). There is a section for 'Phone Type' with two radio buttons: 'use Tone dialing' (selected) and 'use Pulse dialing'. Below this, there are two more checkboxes: 'Ignore Dial-Tone' (checked) and 'Modem is configured to Auto Answer' (unchecked). At the bottom, there are 'OK' and 'Cancel' buttons.

Enable modem Support - Check this box when you want the system to use modem-to-modem connectivity. Note: This option disables some functions in the main setup dialogue.

Modem Init String - Type in a non-default connection string in here ... else the system will setup the modem with the following string "ATE0&D2&C1" Also ATX3 Will be issued if "ignore Dial - Tone" is requested.

Dial Number - The slave teleRanger telephone number. Use ',' to pause etc.

Connect Timeout - The timeout period (in milliseconds) to wait for a connection, that is, the time till carrier is detected.

Inactivity Timeout - The timeout period (in milliseconds) to wait on and idle/silent port before hanging up automatically.

Guarantee max time - The timeout period (in seconds) to guarantee that a control to another device not on the same telephone line will wait before being serviced via a forced hangup ... even if there are still outstanding messages for the currently connected device.

Phone type Tone or pulse phone type.

Ignore Dial- Tone - Used where phone voltages are not correct ... note ATX3 is then sent during connection.

Modem is configured for Auto answer - Default: checked, this is used when you don't want the driver to send ATA's when a "RING" is detected, make sure that ATS0=n (where n > 0) to enable modem side auto answering.

3.3. Master/Standby

When the Adroit Agent server changes between modes (only if configured as a cluster server), the driver is notified, and will not perform any "IR" nor "LT" transmissions when in standby. The input data from stations is still processed as normal but is ignored by the Adroit Server. Any queued commands (or retries) when the server changes from master to standby will still be sent and are not immediately cleared or removed at that time.

4. Supported Addresses

4.1. Modem Configuration Strings (x2 command set)

Tested with US-Robotics 56k modems, compatible with x2 command set.
See Appendix A for additional modem topics.

This string is sent whenever Adroit attempt to communicate to a device, it is also sent at startup of the agent server, and also if your stop/restart your device.

Try to use Factory default settings if possible; AT&F

To use Auto answer mode on the modem: AtS0=n (where n = number of rings)

To not use Auto Answer, make sure you uncheck the AutoAnswer checkbox in the Modem Dialogue.
Make sure AtS0=0 in your string.

Make sure &D is 2, the driver needs normal DTR signals to do its hang-ups etc

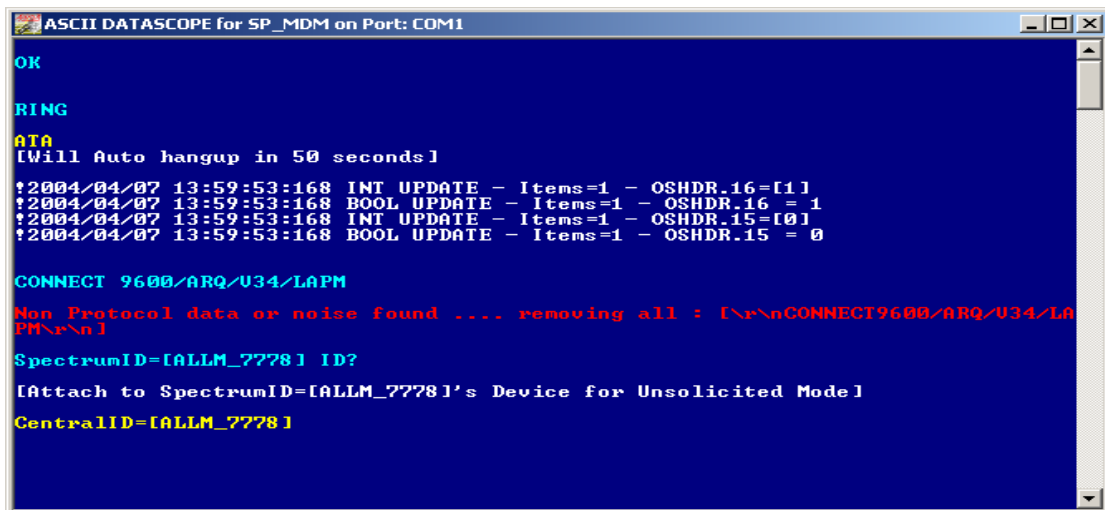
Make sure &C is 1, the driver needs normal Carrier detection to operate correctly.

4.1.1. Example Connection String:

[At&f&w0&w1](#) in "terminal" to reset to factory defaults and save profiles or you could allow adroit to do it at startup.

[ATE0&D2&C1S0=2&W0&W1](#)

Echo off | normal dtr | Normal cd | AA after 2 rings | Save profile to 0 and 1.



```

ASCII DATASCOPE for SP_MDM on Port: COM1
OK
RING
ATA
[Will Auto hangup in 50 seconds]
?2004/04/07 13:59:53:168 INT UPDATE - Items=1 - OSHDR.16=[1]
?2004/04/07 13:59:53:168 BOOL UPDATE - Items=1 - OSHDR.16 = 1
?2004/04/07 13:59:53:168 INT UPDATE - Items=1 - OSHDR.15=[0]
?2004/04/07 13:59:53:168 BOOL UPDATE - Items=1 - OSHDR.15 = 0
CONNECT 9600/ARQ/U34/LAPM
Non Protocol data or noise found .... removing all : [\r\nCONNECT9600/ARQ/U34/LA
PM\r\n]
SpectrumID=[ALLM_7778] ID?
[Attach to SpectrumID=[ALLM_7778]'s Device for Unsolicited Mode]
CentralID=[ALLM_7778]
  
```

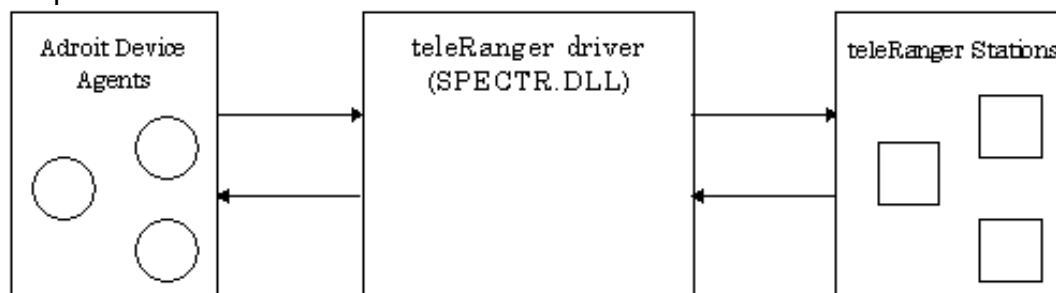
The Non Protocol or noise message above is **not** fatal, some modem handshake may be found to be rejected as above ... ok The "\r\n" etc are just the carriage return line feed converted to printable form they are not actually "\r" they are non-printable and thus not detectable on the scope otherwise!

4.2. Basic Structure of Driver

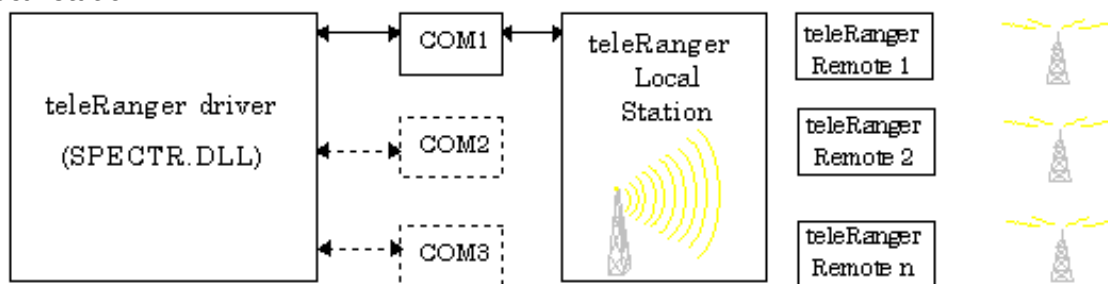
Being a Radio Telemetry System, teleRanger poses some special protocol driver development and configuration challenges:

- Delayed response. With PLC systems a read request is honoured immediately, whereas with a radio telemetry system, several different requests can be sent out before a response to the first one is received. Also, the sequence in which responses arrive is totally unpredictable
- Exception-based. Instead of using a poll-response type protocol like most other Adroit protocol drivers, the teleRanger driver operates in an exception- or event-based mode. This means that messages arrive from the teleRanger equipment in an unsolicited fashion
- Time-tagged. The teleRanger protocol includes the possibility of tagging changes of state and other events with the time of occurrence. The protocol driver needs to be aware of time-tagged information and make sure that this time, as opposed to the current system time, is associated with the event
- Even though Adroit directly connects to only a local teleRanger station, it must maintain communications, etc., on a per remote station basis. This is to prevent all stations being considered bad when only one station may be experiencing communication problems

Given all the above, the following diagram illustrates the context within which the teleRanger protocol driver operates:



From the above diagram it is clear that there is a direct correspondence between teleRanger stations and Adroit device agents, i.e. a separate device agent is configured for each teleRanger outstation. The following diagram shows how the protocol driver connects via an RS232 link to the teleRanger local station



The local station connects via a radio network to its remote outstations. One can see on the above diagram, shown in dotted, that it is possible for different instances of the teleRanger driver to use different COM ports to other local stations that are part of completely separate radio telemetry systems, if necessary.

4.3. Log Retrieval

For datalog retrieval from the RTU via the tag OSCTL.7, or Autolog retrieval...

An Int/real tag must be scanned from OSLOG.1 to hold the count of remaining records.

A Digital tag can be scanned to OSCTL.7 if a manual log retrieval is required.

An Int/real/marshal tag must be scanned to OSHDRB.1 these items all update as a result of a SL, GL or in the case of a Teleflex an IR, which provide an LS.

The OSLOG tag **MUST** be scanned else the retrieval will not be done.

Autologs **DO NOT INITIATE DIALS** while in modem mode.

Autolog retrieval is **NOT ALLOWED DURING UNSOLICITED DIALUP SESSIONS** (i.e. Rtu dialed into the master even if another IR was sent with the subsequent LS , the GL's will not go out... only when the master dials the RTU will the getlog function operate) **EVEN THE MANUAL GETLOG CONTROL BIT WILL NOT BE ALLOWED**

4.4. Addressing Notation

There are four basic items that need to be defined to be able to uniquely identify any point in a teleRanger radio telemetry system. These are:

- Station Address - a number (01 - 99) that defines the particular teleRanger station
- Encode/Decode Channel - a number (00 - 99) that defines the particular I/O module. Inputs are "encoded" with their channel number as part of message transmission, and outputs are "decoded" by teleRanger stations before being recognized ¹
- Module - as per above table
- Bit/Channel Number - a number (1 - 32) that defines the particular input or output on a specific I/O module ²

Note that, teleRanger stations are mapped directly to Adroit devices. This means that the station address field above does not need to form part the Adroit teleRanger addressing notation.

The addressing notation is therefore:

<Module >.[<Encode/Decode Channel>].<Bit/Channel Number>

Note that the Encode/Decode channel field is not always required. Specifically, for outstation header items, which do not relate directly to I/O points in the field, but rather to DIP switch settings, etc., on the outstation, the concept of an encode/decode channel is meaningless and is hence not required. A bit number is often supplied where not necessary and will be 1.

Some examples:

DIO88.04.1[0]	Input No. 1 of 8-channel digital input/ Output card with encode channel set to 04 [0..15 pulse factor-Digitals Outputs Only]
DIN32.04.1	Input No. 1 of 32-channel digital input card with encode channel set to 04
AIN412.7.4	Input No. 4 of 4-channel analog input card with encode channel set to 07
OSHDR.1	Outstation header input no. 1 (master/slave mode)
OSCTL.4	Outstation header control no. 4 (get dip switch settings)

¹ Note that for the dummy counter reset and counter sync modules the encode/decode channel of the relevant CIN4 module should be supplied

² Note that the maximum bit/channel no. is dependent on the module type, for example: 32 for a DIN32 module, and 2 for a AOU28 module.

OSCTL Items

- TX interrogation request
- TX Test transmission
- TX Update request
- TX get dips
- Time Sync
- Hang Modem *(modem mode only)
- Get logs
- Get Outstation time (used in conjunction with OSTIME)

OSHDR Items

- Master mode
- Analog COS
- Digital COS
- Timer A
- Timer B
- Timer C
- TX Ack
- TX Double
- DIP software setting 1
- If DIP is transmitted on change

If IA received

If WR received

If PR received

If TT received

Modem dialling now!

Modem connected (data mode)

OSHDRB Items

- Log feature unavailable on the RTU
- Digitals are enabled for logging
- Analogs are enabled for logging
- Error (=0 reserved for future use)
- Completed (will be 1 if # records in RTU=0)
- RTU logging is set to automatic via dips

OSLOG

Records in RTU log

OSTIME

Outstation Time stored here (1996-01-01-01:01:01:01 = x)

4.5. TeleRanger Module Types and Commands

The following table describes the different teleRanger module types, associated protocol commands, data types, etc.

Type	Module	Description	Protocol Command(s)	Data Type
-	OSHDR	Outstation header input items	WS, PR, WR, TT, IA, modem ind.	Boolean input
-	OSCTL	Outstation header control items	IR, RT, UP, GW, SM, modem cont.	Boolean output
-	OSLOG	# records in RTU	LS	Integer input
3	DIO88	8-channel digital input/output	DI, DC, BP,DT	Boolean input/output
-	DIO88N	DIO8/8 with <u>NO</u> acknowledge	DI, DN, BP,DT	Boolean input/output
4	AIN88	8-channel, 8-bit analog input	AN, AI, AC,AT	8-bit integer input
5	AIN412	4-channel, 12-bit analog input	AN, AI, AC,	16-bit integer input
6	AOU28	2-channel, 8-bit analog output	AC	8-bit integer output
7	AOU212	2-channel, 12-bit analog output	AC	16-bit integer output
8	CIN4	4-channel 16-bit counter	CN, CI, CC, CS, CR	16-bit integer input
-	CRS4	4-channel 16-bit counter reset	GR	Boolean output
-	CZR4	4-channel 16-bit counter sync	CS, CR	Boolean input
9	DIN32	32-channel digital input	DN, DI, DC	Boolean input
-	DOU32	32 bit digital output	DC	Integer Output

Where a type code is missing for a module it is because the module mimics another module or is specially derived.

The DIO88N for example is based on the standard DIO88 but uses a DN instead of a DC command to send outputs, the DN command does not elicit acknowledgements, and is thus preferable in certain cases to a DC command.

The DOU32 is another custom module, it uses a DC command, which is not acknowledged by the RTU.

5. Protocol Definition and Considerations

The following protocol document outlines the subset of the Tele-ranger protocol supported and used by the Adroit Tele-ranger driver. The outline shows the basic telegram format as a skeleton, and then progresses to fill in the components in the data portion of the telegram. The protocol is ASCII, i.e. using only printable characters, decimal values appear as "30" for the value 30 and consists of the binary characters 33,30. Hex values appear as "0A" which is 10 decimal and consists of the binary characters 30,41.

All values are in ascii decimal except \$\$ which are ascii hex notation.

5.1. Basic Telegram Format

5.2.

Telegrams may omit certain fields when not required. Only some telegram types for instance include the time tag which are the option header delimited by "!" to "#"

Description	Width (in bytes)	Contents (format of field)
Time tag prefix	1	!
Time tag	13	yymmddhnnssfff
Separator	1	X
Checksum	2	\$\$
Message Prefix	1	#
Command Type	1	T/R/U/D/I/M/C/N (See below)
System No	1	0
Source (station)	2	00-99/C1/C2
Destination (station)	2	00-99/C1/C2/AL
Command Word	2	(See below)
Data	n	(Depends on module)
Separator	1	+
Checksum	2	\$\$
Terminator	2	<CR> <LF>

5.3. Fields

5.3.1. Command Type Breakdown

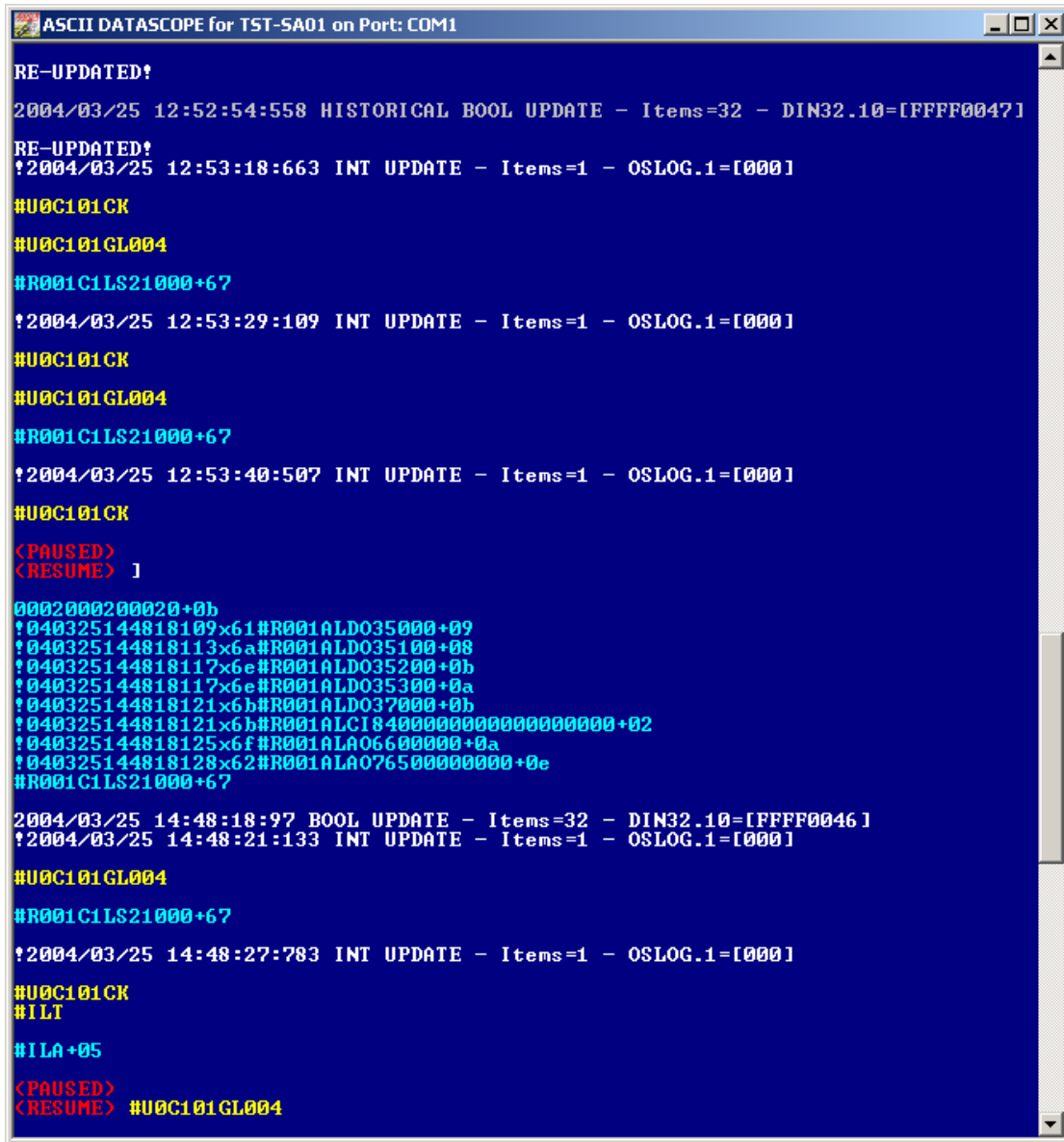
Command	Meaning
R	Received on radio
T	Transmitted on radio
U	Scada "UP" to teleRanger command
D	Tele-ranger "DOWN" to Scada command
I	Internal commands (Scada/Tele-ranger)
M	Message
C	Comment (Tele-Ranger optional information often presented on outstation start-up)
N	NAK (Tele-ranger did not understand command)

5.3.2. Command Words

Command	Meaning
GM	Get time
SM	Set time
MS	Time
PR	Power-on-reset
WR	Watchdog/reset
IN	Initialise outstation
UP	Update request (all I/O is to be sent by the receiving station)
IR	Interrogate request
IA	Interrogate Ack (followed by all I/O)
DN	Digital input data
DI	Interrogation (identical to DN)
DT	Timer mode timeout occurred
DC	Digital change of state
BP	Bit pulse command
GD	Get digital command
DO	Digital output data
AN	Analogue input data
AI	Interrogation (identical to AN)
AT	Timer mode timeout occurred
AC	Analogue change of state
GA	Get Analogue command
AO	Analogue output data
CN	Counter input
CI	Interrogate Counter
CT	Counter timeout occurred
CC	Counter change of state
CS	Counter sync (counter reset by hardware)
GC	Get count
GR	Get and reset
CR	Response to GR
AK	Ack for normal functions such as controls
CK	Ack for GL (Getlogs)
RN/VN	Firmware rev
GW	Get dip sWitches
SW	Set dip sWitch
WS	Switch setting is
GB	Get debounce settings
GP	Get packet
PS	Response (+data)
WP	Write Packet (+data)
PA	Packet Acknowledge

5.4. Datascope Display

All received characters are displayed on datascope; all received nulls are first converted to carriage returns before display. After reception and display, parsing begins.



```

ASCII DATASCOPE for TST-SA01 on Port: COM1

RE-UPDATED!
2004/03/25 12:52:54:558 HISTORICAL BOOL UPDATE - Items=32 - DIN32.10=[FFFF0047]
RE-UPDATED!
!2004/03/25 12:53:18:663 INT UPDATE - Items=1 - OSLOG.1=[000]
#U0C101CK
#U0C101GL004
#R001C1LS21000+67
!2004/03/25 12:53:29:109 INT UPDATE - Items=1 - OSLOG.1=[000]
#U0C101CK
#U0C101GL004
#R001C1LS21000+67
!2004/03/25 12:53:40:507 INT UPDATE - Items=1 - OSLOG.1=[000]
#U0C101CK
<PAUSED>
<RESUME> 1
0002000200020+0b
!040325144818109x61#R001ALD035000+09
!040325144818113x6a#R001ALD035100+08
!040325144818117x6e#R001ALD035200+0b
!040325144818117x6e#R001ALD035300+0a
!040325144818121x6b#R001ALD037000+0b
!040325144818121x6b#R001ALC1840000000000000000+02
!040325144818125x6f#R001ALA066000000+0a
!040325144818128x62#R001ALA076500000000+0e
#R001C1LS21000+67
2004/03/25 14:48:18:97 BOOL UPDATE - Items=32 - DIN32.10=[FFFF0046]
!2004/03/25 14:48:21:133 INT UPDATE - Items=1 - OSLOG.1=[000]
#U0C101GL004
#R001C1LS21000+67
!2004/03/25 14:48:27:783 INT UPDATE - Items=1 - OSLOG.1=[000]
#U0C101CK
#ILT
#ILA+05
<PAUSED>
<RESUME> #U0C101GL004
  
```

5.5. Advanced Registry Features: (rev 41 and above)

As a rule of thumb , if installing a newer driver or where you know you will need to insert one of the following , open each device in the configurator (psexec.exe) and hit ok , this will resave the registry, and add any keys if necessary else make sure you type in the following precisely as it is shown here. No special cases yet...

6. General Notes and Troubleshooting

6.1. Contact Details

Email: support@adroit.co.za and drivers@adroit.co.za

Phone: +27 11 658-8100

Web: www.adroit.co.za

6.2. Adroit Technologies Knowledge Base

For additional information, please refer to our Knowledgebase website:

[Latest Drivers topics - Features, discussions, tips, tricks, questions, problems and feedback \(adroittechnologiesautomation.com\)](http://adroittechnologiesautomation.com)

7. Appendix A – How to Troubleshoot and Setup Modems.

This discussion describes landline dialup modem support , but gsm modems are also compatible to the basic command sets so there is no reason for them not to work either.

Where applicable I will refer to the Hayes (x.2) command set as well as the Duxbury command set.

(*h) – X2 Hayes only (e.g. US robotics)

(*d) – Duxbury only.

Nothing means identical for both command sets.

7.1. 1st Steps

Use HyperTerminal to establish initial communications with the modems , make sure you can communicate with the modem by creating a hyper terminal session manually to the appropriate com port. Type in any command and check whether the modem acknowledges it. Try AT(return) you should see OK. [ATZ, ATI4(*h) or AT&V(*d) for current config info etc)

If no ack is found try a command that will definitely return something ATI4(*h) AT&V(*d).

If no acks are sent but the about info does get sent check result codes are being sent Q = 0. And the codes must be returned as text V=1.

AT&F will reset the modem factory defaults.

Make sure Carrier detect is enabled &C1. *NB – for driver hang up ability.

Make sure DTR control is enabled &D2. *NB – for driver hang up ability.

If you want the modem to auto-answer make sure S0>0 (i.e. S0=2 for 2 rings then answer)

Make sure Echo is off when you are finished testing E=0. *NB – for driver logic.

Make sure B0(*h). Data baud rate will follow connection train rate.

AT&W(*h) will save the profile , older modems have 2 profiles AT&W0 , AT&W1...

7.2. For Remote Modem

Make sure the following is set : S0=0, spectrum answers manually.

&D0 spectrum does not control the DTR line and thus must be disabled/ignored!

Recommended ATE0 s0=0 &D2 &C1&W done manually via terminal! Make sure the save went through by turning off the modem and then doing a ATI4(*h) to check.

7.3. For Master Modem

S0 = 2 , two rings until auto answer, then the device modem "modem is configured for auto answer" must be checked , otherwise if its unchecked S0=0, then the driver will do a manual ATA. (It is recommended to use the modems auto answer feature)

&D2 SPECTR Driver must have control of the DTR line to do proper hangup!
 &C1 Driver must have be able to see if Carrier has been established.
 E0 Echo must be off! The driver cannot connect if echo is on

Recommended connection string ATE0s0=0&D2&C1&W some modems don't like the &W it them must be done manually via terminal!

8. Revisions

8.1. Document Revision

Rev.	Date	Author	Comment	Approval
1.0	14-Dec-2022	J. Duncan	New Format	D. Jordaan
1.1		K. Robinson	Changes to format, updated image resources.	D. Jordaan
2.0				
3.0				
4.0				
5.0				
6.0				

8.2. DLL Revision

Rev.	Date	Changes
48	10/10/03	Aou28,Aou212 Were not unpacking correctly (rejected as noise since 44c) , Job (Scan Agents) now map only one device module per job Added code to retrieve historical data correctly use in conjunction with t_scan.dll >=rev34 from 4.2.13 or >=rev6 from 5.0.3 onwards Added Auto retrieval as well and retry mechanism ... Improved scope output (now include module/channel/subchannel)
50	20/11/03	Auto GL timer added , also fixed GL retries bug created in 48, also found if an event was received first in a batch of other historical data the event timestamp was discarded and current time inserted , this won't be a serious issue if the RTU and master was in sync but fixed anyway!
51h	25/03/04	Stabilisation and bug fixes, Auto GLs modified to use ACs.
51o	29/04/04	Unsolicited crash fix.
59	29/04/04	Latency now correctly applied separately to each coms resource (central station).
65	28-Sep 2018	Fixed missing performance counters when Agent Server is running as a service.

